

Lab - Exploring Processes, Threads, Handles, and Windows Registry (DAYUTH THY)

Objectives

In this lab, you will explore the processes, threads, and handles using Process Explorer in the SysInternals Suite. You will also use the Windows Registry to change a setting.

Part 1: Exploring Processes

Part 2: Exploring Threads and Handles

Part 3: Exploring Windows Registry

Required Resources

- 1 Windows PC with internet access

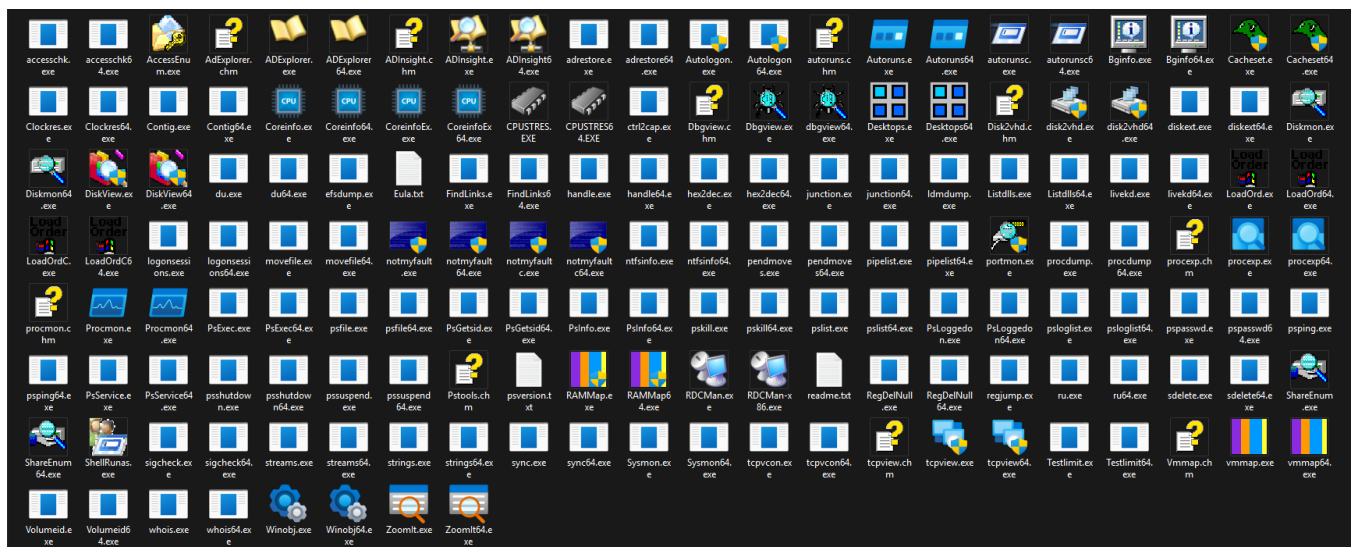
Instructions

Part 1: Exploring Processes

In this part, you will explore processes. Processes are programs or applications in execution. You will explore the processes using Process Explorer in the Windows SysInternals Suite. You will also start and observe a new process.

Step 1: Download Windows SysInternals Suite.

- Navigate to the following link to download Windows SysInternals Suite:
<https://technet.microsoft.com/en-us/sysinternals/bb842062.aspx>
- After the download is completed, extract the files from the folder.
- Leave the web browser open for the following steps.



Step 2: Explore an active process.

- a. Navigate to the SysinternalsSuite folder with all the extracted files.
- b. Open **procexp.exe**. Accept the Process Explorer License Agreement when prompted.
- c. The Process Explorer displays a list of currently active processes.

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- d. To locate the web browser process, drag the **Find Window's Process** icon into the opened web browser window. Microsoft Edge was used in this example.

The screenshot displays the Process Explorer window from Sysinternals, showing a list of running processes. The 'Process' column lists various applications, including multiple instances of 'msedge.exe' (Microsoft Edge) and 'firefox.exe' (Firefox). The 'CPU' column shows the percentage of CPU usage for each process. The 'Private Bytes' and 'Working Set' columns show the memory usage. The 'PID' column shows the process ID. The 'Description' column shows the name of the process. The 'Company Name' column shows the company that developed the process. The 'Virus Total' column shows the virus scan results.

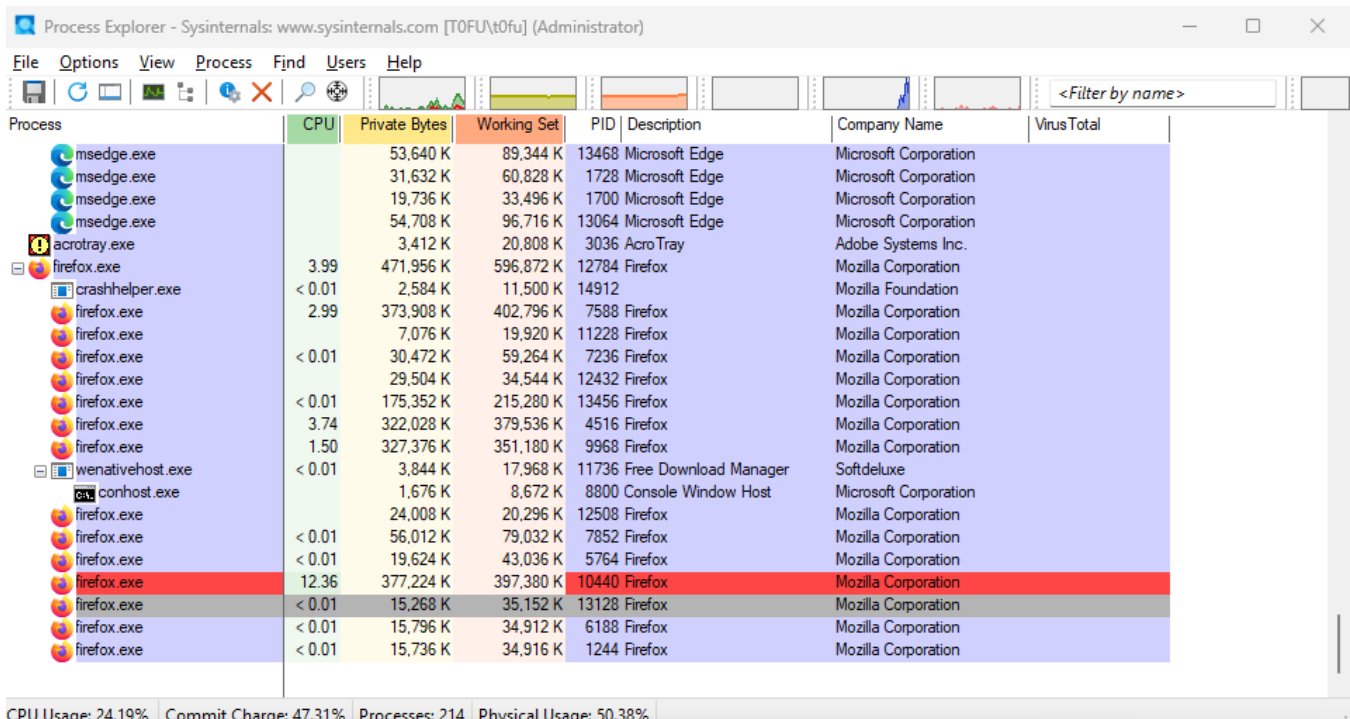
Below the main process list, there is a 'Process Explorer Search' dialog box. The 'Handle or DLL substring' field is set to 'firefox'. The 'Search' button is highlighted. The search results are displayed in a table with the following columns: Process, PID, Type, and Name. The results show various instances of 'firefox.exe' and its associated DLLs and threads.

Process	PID	Type	Name
svchost.exe	1352	Process	firefox.exe(7236)
svchost.exe	1352	Process	firefox.exe(7852)
svchost.exe	1352	Process	firefox.exe(12784)
svchost.exe	1352	Process	firefox.exe(11600)
svchost.exe	1352	Process	firefox.exe(13456)
svchost.exe	1352	Process	firefox.exe(9968)
svchost.exe	1352	Process	firefox.exe(4516)
svchost.exe	1352	Process	firefox.exe(12508)
svchost.exe	1352	Process	firefox.exe(7588)
svchost.exe	1352	Process	firefox.exe(11652)
svchost.exe	1352	Process	firefox.exe(13784)
svchost.exe	1352	Process	firefox.exe(5764)
svchost.exe	1352	Process	firefox.exe(3176)
svchost.exe	1352	Process	firefox.exe(12432)
svchost.exe	2464	Process	firefox.exe(12784)
svchost.exe	2464	Process	firefox.exe(7588)
svchost.exe	2912	Process	firefox.exe(12784)
firefox.exe	3176	DLL	C:\Program Files\Mozilla Firefox\firefox.exe
firefox.exe	3176	DLL	C:\Program Files\Mozilla Firefox\xul.dll
firefox.exe	3176	DLL	C:\Program Files\Mozilla Firefox\gkcodecs.dll
firefox.exe	3176	DLL	C:\Program Files\Mozilla Firefox\nss3.dll
firefox.exe	3176	DLL	C:\Program Files\Mozilla Firefox\mozglue.dll
firefox.exe	3176	DLL	C:\Program Files\Mozilla Firefox\lgpllibs.dll
firefox.exe	3176	File	C:\Program Files\Mozilla Firefox
firefox.exe	3176	File	C:\Program Files\Mozilla Firefox
firefox.exe	3176	Thread	firefox.exe(3176): 13516
firefox.exe	3176	Thread	firefox.exe(3176): 13516
firefox.exe	3176	Thread	firefox.exe(3176): 11360
firefox.exe	3176	Thread	firefox.exe(3176): 13516
firefox.exe	3176	Thread	firefox.exe(3176): 13516
firefox.exe	3176	Thread	firefox.exe(3176): 13516
firefox.exe	3176	Thread	firefox.exe(3176): 12044
firefox.exe	3176	Thread	firefox.exe(3176): 11360
firefox.exe	3176	Thread	firefox.exe(3176): 11360
firefox.exe	3176	Thread	firefox.exe(3176): 6668
firefox.exe	3176	Thread	firefox.exe(3176): 13516
firefox.exe	3176	Thread	firefox.exe(3176): 1828
firefox.exe	3176	Thread	firefox.exe(3176): 2852
firefox.exe	3176	Thread	firefox.exe(3176): 2852

1316 matching items.

The Microsoft Edge process can be terminated in the Process Explorer. Right-click the selected process and select **Kill Process**. Click **OK** to continue.

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Process	CPU	Private Bytes	Working Set	PID	Description	Company Name	Virus Total
msedge.exe		53,640 K	89,344 K	13468	Microsoft Edge	Microsoft Corporation	
msedge.exe		31,632 K	60,828 K	1728	Microsoft Edge	Microsoft Corporation	
msedge.exe		19,736 K	33,496 K	1700	Microsoft Edge	Microsoft Corporation	
msedge.exe		54,708 K	96,716 K	13064	Microsoft Edge	Microsoft Corporation	
acrotray.exe		3,412 K	20,808 K	3036	Acro Tray	Adobe Systems Inc.	
firefox.exe	3.99	471,956 K	596,872 K	12784	Firefox	Mozilla Corporation	
crashhelper.exe	< 0.01	2,584 K	11,500 K	14912		Mozilla Foundation	
firefox.exe	2.99	373,908 K	402,796 K	7588	Firefox	Mozilla Corporation	
firefox.exe		7,076 K	19,920 K	11228	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	30,472 K	59,264 K	7236	Firefox	Mozilla Corporation	
firefox.exe		29,504 K	34,544 K	12432	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	175,352 K	215,280 K	13456	Firefox	Mozilla Corporation	
firefox.exe	3.74	322,028 K	379,536 K	4516	Firefox	Mozilla Corporation	
firefox.exe	1.50	327,376 K	351,180 K	9968	Firefox	Mozilla Corporation	
wenativehost.exe	< 0.01	3,844 K	17,968 K	11736	Free Download Manager	Softdeluxe	
conhost.exe		1,676 K	8,672 K	8800	Console Window Host	Microsoft Corporation	
firefox.exe		24,008 K	20,296 K	12508	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	56,012 K	79,032 K	7852	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	19,624 K	43,036 K	5764	Firefox	Mozilla Corporation	
firefox.exe	12.36	377,224 K	397,380 K	10440	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	15,268 K	35,152 K	13128	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	15,796 K	34,912 K	6188	Firefox	Mozilla Corporation	
firefox.exe	< 0.01	15,736 K	34,916 K	1244	Firefox	Mozilla Corporation	

CPU Usage: 24.19% Commit Charge: 47.31% Processes: 214 Physical Usage: 50.38%

What happened to the web browser window when the process is killed?

Answer: Gone

Step 3: Start another process.

- Open a Command Prompt. (**Start** > search **Command Prompt** > select **Command Prompt**)
- Drag the **Find Window's Process** icon into the Command Prompt window and locate the highlighted Command Prompt process in Process Explorer.
- The process for the Command Prompt is cmd.exe. Its parent process is explorer.exe process. The cmd.exe has a child process, conhost.exe.
- Navigate to the Command Prompt window. Start a ping at the prompt and observe the changes under the cmd.exe process.

What happened during the ping process?

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The screenshot displays two windows from a Windows system. The top window is 'Process Explorer - Sysinternals: www.sysinternals.com [T0FU\t0fu] (Administrator)'. It shows a list of running processes with columns for CPU, Private Bytes, Working Set, PID, Description, Company Name, and VirusTotal. The 'explorer.exe' process is highlighted in blue, and 'conhost.exe' is highlighted in green. The bottom window is a 'Command Prompt - ping gox' showing the results of a ping command to google.com. The output indicates that the ping was successful with 0% loss and an average round trip time of 26ms.

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name	VirusTotal
csrss.exe	< 0.01	2,960 K	7,524 K	1016			
winlogon.exe		2,736 K	17,368 K	928	Windows Logon Application	Microsoft Corporation	
fontdrvhost.exe		6,220 K	14,016 K	1260	Usemode Font Driver Host	Microsoft Corporation	
dwm.exe	0.25	217,556 K	248,404 K	1476	Desktop Window Manager	Microsoft Corporation	
explorer.exe	< 0.01	251,448 K	417,984 K	7388	Windows Explorer	Microsoft Corporation	
SecurityHealthSystray.exe		2,304 K	14,020 K	11864	Windows Security notificatio...	Microsoft Corporation	
RtkAudUService64.exe	< 0.01	4,812 K	18,936 K	11944	Realtek HD Audio Universal ...	Realtek Semiconductor	
OneDrive.exe	< 0.01	75,584 K	146,204 K	4632	Microsoft OneDrive	Microsoft Corporation	
WINWORD.EXE	< 0.01	186,420 K	352,688 K	12936	Microsoft Word	Microsoft Corporation	
aimgr.exe		4,628 K	24,352 K	7216	Local AI Manager for Micros...	Microsoft Corporation	
ai.exe	< 0.01	22,312 K	40,636 K	5516	Local AI Host for Microsoft 3...	Microsoft Corporation	
ai.exe	< 0.01	55,604 K	75,420 K	11144	Local AI Host for Microsoft 3...	Microsoft Corporation	
ai.exe	< 0.01	46,420 K	58,588 K	10580	Local AI Host for Microsoft 3...	Microsoft Corporation	
procexp64.exe	0.76	46,512 K	84,044 K	1488	Sysinternals Process Explorer	Sysinternals - www.sysinter...	
cmd.exe	< 0.01	4,012 K	8,980 K	5872	Windows Command Processor	Microsoft Corporation	
conhost.exe	< 0.01	2,032 K	11,348 K	13724	Console Window Host	Microsoft Corporation	
PING.EXE	0.51	1,944 K	9,560 K	6420	TCP/IP Ping Command	Microsoft Corporation	
msedge.exe	< 0.01	87,760 K	193,596 K	15336	Microsoft Edge	Microsoft Corporation	
msedge.exe		2,888 K	11,208 K	6208	Microsoft Edge	Microsoft Corporation	
msedge.exe		32,556 K	51,944 K	4060	Microsoft Edge	Microsoft Corporation	
msedge.exe		15,912 K	44,460 K	10036	Microsoft Edge	Microsoft Corporation	
msedge.exe		8,824 K	19,616 K	9116	Microsoft Edge	Microsoft Corporation	
msedge.exe	< 0.01	31,616 K	55,048 K	14052	Microsoft Edge	Microsoft Corporation	
msedge.exe		53,640 K	89,340 K	13468	Microsoft Edge	Microsoft Corporation	
msedge.exe		31,636 K	60,832 K	1728	Microsoft Edge	Microsoft Corporation	

```
C:\Users\t0fu>ping google.com

Ping statistics for 172.217.25.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 28ms, Average = 26ms

C:\Users\t0fu>ping google.com

Pinging google.com [172.217.25.46] with 32 bytes of data:
Reply from 172.217.25.46: bytes=32 time=26ms TTL=116
Reply from 172.217.25.46: bytes=32 time=25ms TTL=116
Reply from 172.217.25.46: bytes=32 time=27ms TTL=116
Reply from 172.217.25.46: bytes=32 time=25ms TTL=116
```

- e. As you review the list of active processes, you find that the child process conhost.exe may be suspicious. To check for malicious content, right-click **conhost.exe** and select **Check VirusTotal**. When prompted, click **Yes** to agree to VirusTotal Terms of Service. Then click **OK** for the next prompt.
- f. Expand the Process Explorer window or scroll to the right until you see the VirusTotal column. Click the link under the VirusTotal column. The default web browser opens with the results regarding the malicious content of conhost.exe.

Answer: No Virus Detected

- g. Right-click the cmd.exe process and select **Kill Process**.
What happened to the child conhost.exe?

Answer: Gone Too as Well

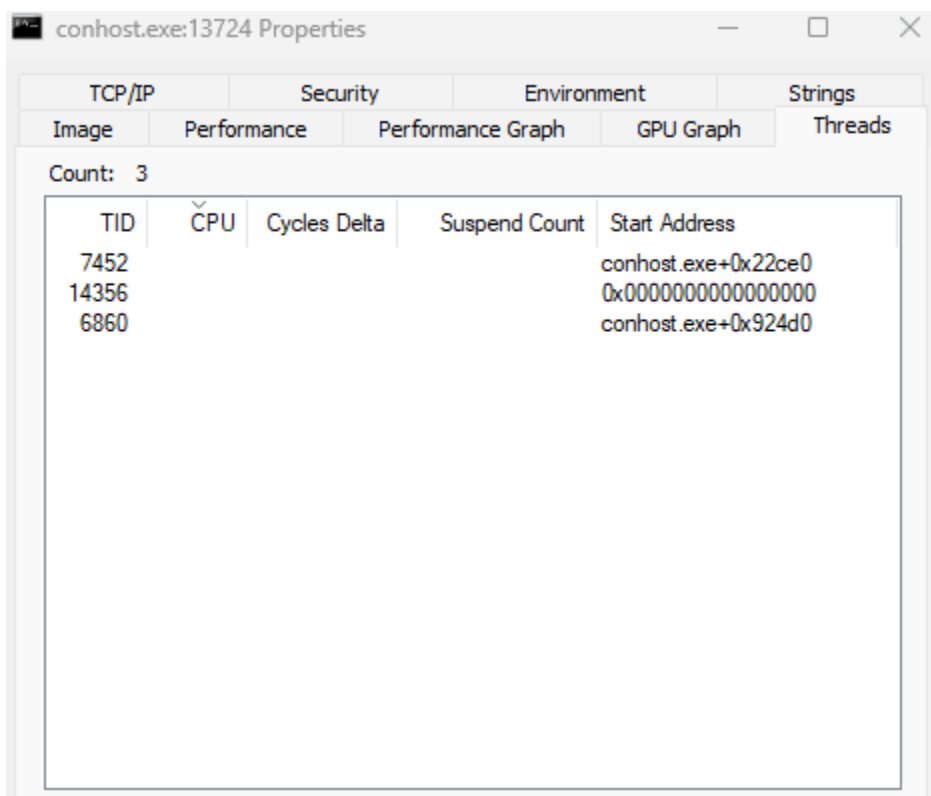
Part 2: Exploring Threads and Handles

In this part, you will explore threads and handles. Processes have one or more threads. A thread is a unit of execution in a process. A handle is an abstract reference to memory blocks or objects managed by an operating system. You will use Process Explorer (procxp.exe) in Windows SysInternals Suite to explore the threads and handles.

Step 1: Explore threads.

- Open a command prompt.
- In Process Explorer window, right-click conhost.exe and Select **Properties**..... Click the **Threads** tab to view the active threads for the conhost.exe process. Click **OK** to continue if prompted by a warning dialog box.
- Examine the details of the thread.

What type of information is available in the Properties window?



- Click **OK** to continue.

Step 2: Explore handles.

- In the Process Explorer, click **View** > select **Lower Pane View** > **Handles** to view the handles associated with the conhost.exe process.

Examine the handles. What are the handles pointing to?

The screenshot shows Process Explorer with the 'Handles' view selected. The top pane lists processes, and the bottom pane shows the handles for the selected process, conhost.exe.

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name	Virus Total
WINWORD.EXE	< 0.01	204,896 K	370,152 K	12936	Microsoft Word	Microsoft Corporation	
aimgr.exe	< 0.01	4,628 K	24,348 K	7216	Local AI Manager for Micros...	Microsoft Corporation	
ai.exe	< 0.01	22,352 K	40,660 K	5516	Local AI Host for Microsoft 3...	Microsoft Corporation	
ai.exe	< 0.01	58,624 K	77,972 K	11144	Local AI Host for Microsoft 3...	Microsoft Corporation	
ai.exe	< 0.01	48,208 K	61,296 K	10580	Local AI Host for Microsoft 3...	Microsoft Corporation	
cmd.exe		3,824 K	9,188 K	5872	Windows Command Processor	Microsoft Corporation	
conhost.exe		1,864 K	11,412 K	13724	Console Window Host	Microsoft Corporation	
procexp64.exe	0.99	33,240 K	66,472 K	6316	Sysinternals Process Explorer	Sysinternals - www.sysinter...	
ShippingTool.exe	2.23	14,164 K	48,812 K	12984			
msedge.exe	< 0.01	87,752 K	147,504 K	15336	Microsoft Edge	Microsoft Corporation	
msedge.exe		2,888 K	10,660 K	6208	Microsoft Edge	Microsoft Corporation	
msedge.exe		22,556 K	40,669 K	4060	Microsoft Edge	Microsoft Corporation	

Name	Description	Company Name	Path
advapi32.dll	Advanced Windows 32 Base API	Microsoft Corporation	C:\Windows\System32\advapi32.dll
C_1252.NLS			C:\Windows\System32\C_1252.NLS
C_1252.NLS			C:\Windows\System32\C_1252.NLS
C_437.NLS			C:\Windows\System32\C_437.NLS
C_850.NLS			C:\Windows\System32\C_850.NLS
C_850.NLS			C:\Windows\System32\C_850.NLS
cmd.exe	Windows Command Processor	Microsoft Corporation	C:\Windows\System32\cmd.exe
cmd.exe.mui	Windows Command Processor	Microsoft Corporation	C:\Program Files\WindowsApps\Microsoft.LanguageExperi...
combase.dll	Microsoft COM for Windows	Microsoft Corporation	C:\Windows\System32\combase.dll
comctl32.dll	User Experience Controls Library	Microsoft Corporation	C:\Windows\WinSxS\amd64_microsoft.windows.common-c...

CPU Usage: 12.63% Commit Charge: 37.80% Processes: 209 Physical Usage: 37.59%

- Close the Process Explorer when finished.

Part 3: Exploring Windows Registry

The Windows Registry is a hierarchical database that stores most of the operating systems and desktop environment configuration settings.

- To access the Windows Registry, click **Start** > Search for **regedit** and select **Registry Editor**. Click **Yes** when asked to allow this app to make changes.

The Registry Editor has five hives. These hives are at the top level of the registry.

- HKEY_CLASSES_ROOT is actually the Classes subkey of HKEY_LOCAL_MACHINE\Software\ . It stores information used by registered applications like file extension association, as well as a programmatic identifier (ProgID), Class ID (CLSID), and Interface ID (IID) data.
- HKEY_CURRENT_USER contains the settings and configurations for the users who are currently logged in.
- HKEY_LOCAL_MACHINE stores configuration information specific to the local computer.
- HKEY_USERS contains the settings and configurations for all the users on the local computer. HKEY_CURRENT_USER is a subkey of HKEY_USERS.
- HKEY_CURRENT_CONFIG stores the hardware information that is used at startup by the local computer.

- In a previous step, you had accepted the EULA for Process Explorer. Navigate to the EulaAccepted registry key for Process Explorer.

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Click to select Process Explorer in **HKEY_CURRENT_USER > Software > Sysinternals > Process Explorer**. Scroll down to locate the key **EulaAccepted**. Currently, the value for the registry key EulaAccepted is 0x00000001(1).

- Double-click **EulaAccepted** registry key. Currently the value data is set to 1. The value of 1 indicates that the EULA has been accepted by the user.
- Change the **1** to **0** for Value data. The value of 0 indicates that the EULA was not accepted. Click **OK** to continue.

What is value for this registry key in the Data column?

Divider	REG_BINARY	00 00 00 00 00 00 e0 3f
DIIColumnCount	REG_DWORD	0x00000004 (4)
DIIPropWindow...	REG_BINARY	2c 00 00 00 00 00 00 00 00 00 00 00
DIISortColumn	REG_DWORD	0x00000000 (0)
DIISortDirection	REG_DWORD	0x00000001 (1)
ETWstandardUs...	REG_DWORD	0x00000000 (0)
EulaAccepted	REG_DWORD	0x00000000 (0)
FindWindowpla...	REG_BINARY	2c 00 00 00 00 00 00 00 01 00 00 00 ff f
FormatIoBytes	REG_DWORD	0x00000001 (1)
GpuNodeUsage...	REG_DWORD	0x00000001 (1)
GpuNodeUsage...	REG_DWORD	0x00000000 (0)
HandleColumn...	REG_DWORD	0x00000002 (2)
HandleSortCol...	REG_DWORD	0x00000000 (0)

- Open the **Process Explorer**. Navigate to the folder where you have downloaded SysInternals. Open the folder **SysInternalsSuite > Open procexp.exe**.

When you open the Process Explorer, what did you see?

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name	Virtual
System	0.25	56 K	6,660 K	4			

CPU Usage: 6.89% | Commit Charge: 37.56% | Processes: 194 | Physical Usage: 36.45%

